

**DSC 288R Project Progress Report - Image-Based
Mushroom Edibility Classification with Deep Learning
Project Group #8**

Joanna Tam j7tam@ucsd.edu	Nicholas Shor nshor@ucsd.edu	Chiu-Chiu (JoJo) Lin jol145@ucsd.edu	Xuewen Yang xuy001@ucsd.edu
-------------------------------------	--	--	---------------------------------------

1	Background	2
2	Dataset name and citation/link	2
3	Data Pipeline	2
4	EDA Description	4
5	Feature Engineering	11
6	Model Selection and Architecture	11
7	Progress Report Details	12
8	Team Member Contribution	13
9	Risks and Mitigation	13
	References	15

1 Background

Mushroom identification is an important problem in ecology, food safety, and environment studies. Correct identification of mushrooms based on their appearance is challenging since many mushrooms share similar visual characteristics, and some mushrooms that look harmless can be toxic or fatal. This will make manual identification difficult, especially for non-experts who lack the knowledge to distinguish mushroom species accurately. The existing mushroom identification tools often require experts' knowledge, which will rely on manual visual comparison and do not scale well to the real world image variability. An image-based learning model can automate the process for identifying mushrooms based on appearance in the large datasets and identify the mushroom edibility.

2 Dataset name and citation/link

We are using the Mushroom Classification Dataset from Kaggle.com: <https://www.kaggle.com/datasets/zedsden/mushroom-classification-dataset>.

3 Data Pipeline

3.1 Purpose

The purpose of the data pipeline is to construct a clean image dataset that can support the robust training and evaluation of the deep learning models for the real-world mushroom images. In addition to basic information, the pipeline will also address the challenges in existing identification tools such as inconsistent image quality, image colors, etc. Standardizing input dataset helps improve model performance.

3.2 Pipeline design and details

1. **Data consolidation:**

- (a) Iterate through all mushroom images from directory structure and retrieve image information like image paths, labels (e.g., edible / poisonous / deadly / conditionally edible), species, image width, height, and resolution.
- (b) Remove corrupted or unreadable images.
- (c) Store this information in a single CSV file - *mushroom_paths.csv*.

2. **Data enrichment:**

- (a) Derive the aspect ratio from the width and height of images.
- (b) Use perceptual hashing to generate an image hash value.
- (c) Add the derived attributes to the dataset CSV.

3. **De-duplication:**

- (a) Identified duplicate and near-duplicate images based on perceptual hash values.
 - (b) Exact duplicates were defined as images sharing the same pHash value, while near-duplicates were identified using Hamming distance between pHash values.
 - (c) For each group of duplicate and near-duplicate images, the image with the highest resolution (largest number of pixels) was retained, as it preserves more visual information and is expected to improve model performance.
 - (d) Metadata entries corresponding to lower-resolution duplicate images were removed from the CSV file - *mushroomDedupExact.csv* and *mushroomDedupNear.csv*
4. **Standardization:**
- (a) Image standardization was applied dynamically during data loading rather than modifying the original image files.
 - (b) All images were resized to a fixed input resolution to ensure consistent input dimensions for convolutional neural networks.
 - (c) Pixel values were scaled and normalized using predefined channel-wise statistics.
 - (d) Standardization was applied consistently across training, validation, and test datasets.
5. **Data Augmentation:**
- (a) Data augmentation was applied only to the training dataset.
 - (b) Augmentation techniques included random horizontal flipping, rotation, and color jittering to simulate natural variations in orientation and lighting.
 - (c) Augmentation was performed on-the-fly during data loading and did not alter the original images.
 - (d) Validation and test datasets were excluded from augmentation to ensure unbiased evaluation.

3.3 Description of outputs

1. A consolidated metadata CSV file containing image file paths, class labels, species information, image width, height, resolution, aspect ratio, and perceptual hash values generated during data consolidation and enrichment.
2. An **exact-deduplicated dataset** produced by removing images with identical perceptual hash (pHash) values. The original dataset has total of **102,711** images. **8,013 (7.8%)** are duplicate, which were removed, and resulting of **94,698** images.
3. A **near-deduplicated dataset** produced by further removing visually similar images identified using Hamming distance between perceptual hash values. It removed **8,080 (7.87%)** images and resulting of **94,631** images.
4. The near-deduplicated dataset consists of 94,631 unique images and was used, which has higher quality image data.

4 EDA Description

4.1 Types of Analysis Used

1. **Univariate Analysis:** Class distribution, image width, height, resolution, and aspect ratio.
2. **Multivariate Analysis:** Relationships between image width and height to identify potential outliers.
3. **Graphical Analysis:** Bar charts, pie charts, histograms, scatter plots, and sample image grids.
4. **Non-graphical Analysis:** Summary statistics and duplicate frequency analysis using perceptual hash values.

4.2 Key Findings from EDA

1. **Basic Quality Checks:**
 - (a) Raw dataset has total of **102,711 images**.
 - (b) No significant missing values are found in the metadata fields (Table 1).

Table 1: Missing values per columns in metadata

	Missing Values
file_path	0
depth	0
file_name	0
file_ext	0
width	0
num_pixels	0
phash	0
aspect_ratio	0
class	0

- (c) All image files used in analysis were verified that are readable and valid.
2. **Class distribution:**
 - (a) The bar chart (Figure 1) and the pie chart (Figure 2) show the class distribution by number and percentage.

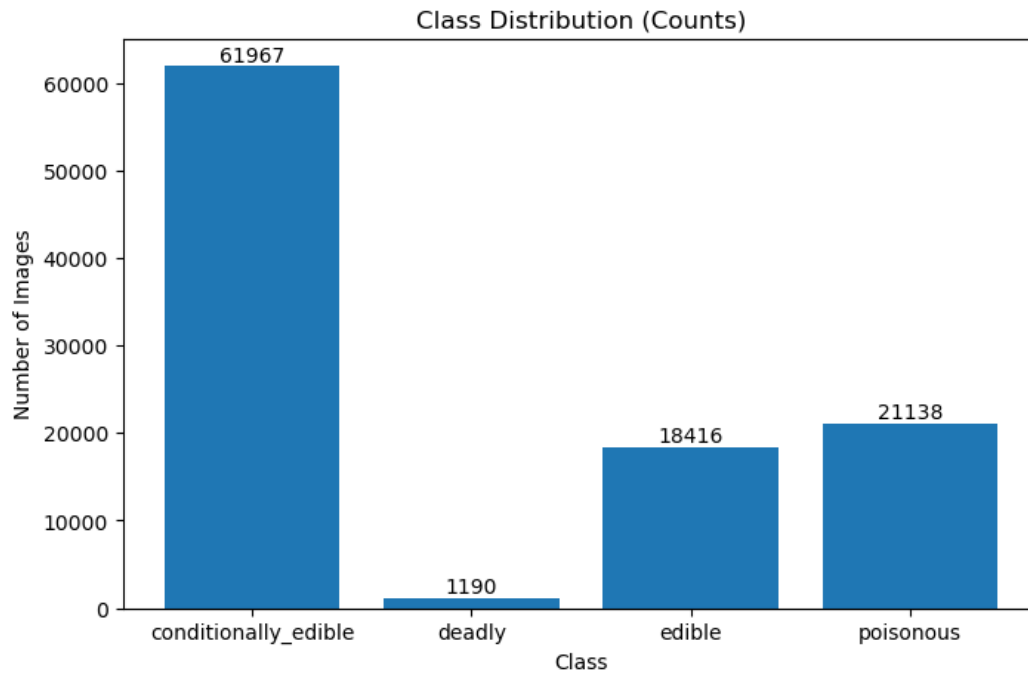


Figure 1: Distribution of Mushroom Classes by Image Count

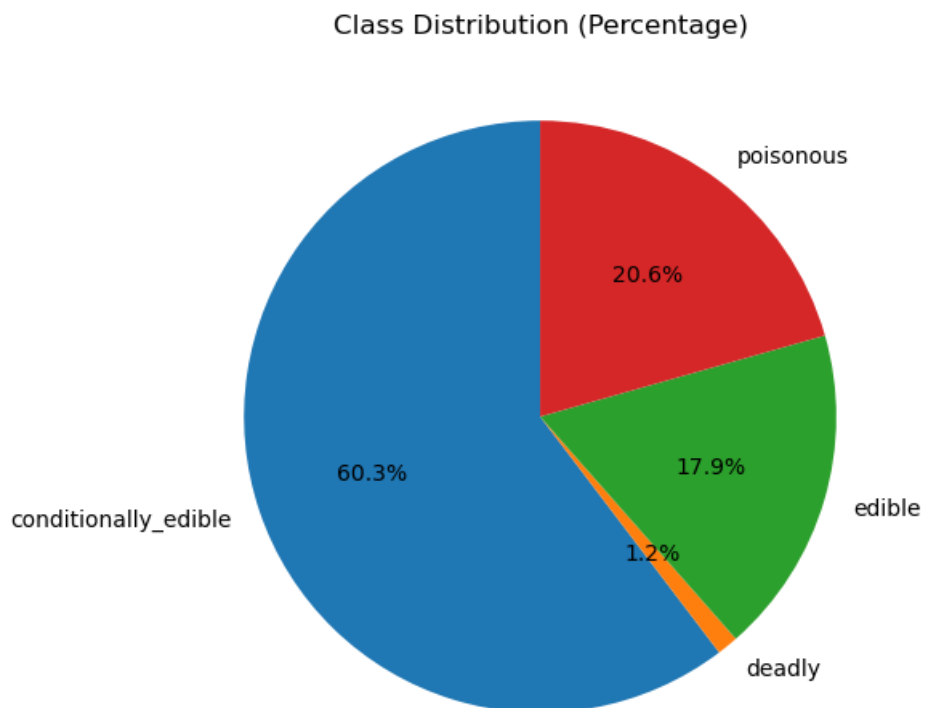


Figure 2: Percentage Distribution of Mushroom Classes

- (b) The result shows the data are imbalanced across classes that conditionally edible dominate in class with 60.3% and deadly class only contain 1.2%.

3. Image Dimension Distribution:

- (a) Image width (Figure 3), height (Figure 4), and resolution (width $\times$ height) (Figure 5) show large variability, indicating a wide range of image sizes across the dataset.

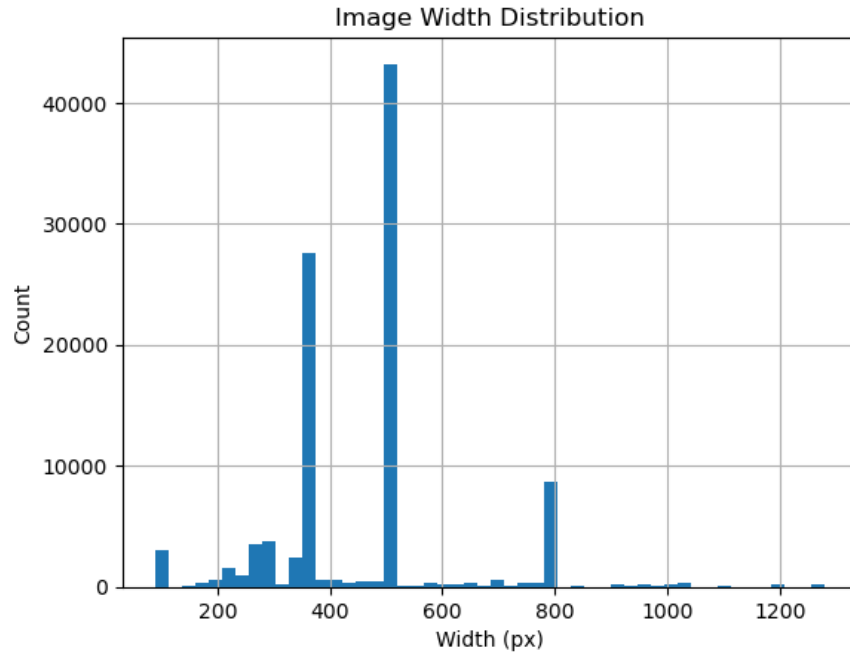


Figure 3: Distribution of Image Width of Mushrooms

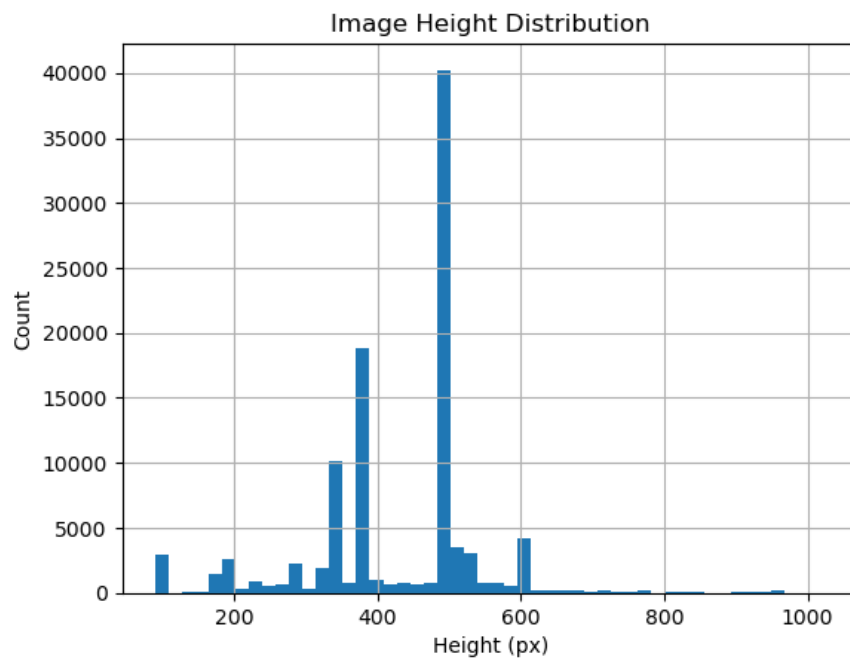


Figure 4: Distribution of Image Height of Mushrooms

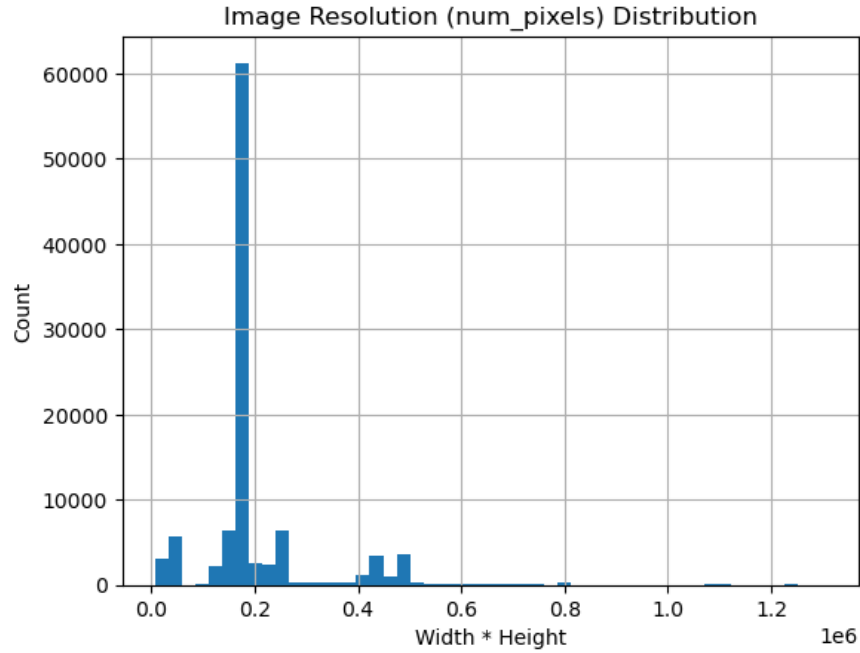


Figure 5: Distribution of Image Resolution of Mushrooms

(b) These support the findings to resize images to have fixed input resolution in standardization.

4. Aspect Ratio Analysis:

(a) Histograms of aspect ratio distribution (Figure 6) showed that most of the images fall within a reasonable range.

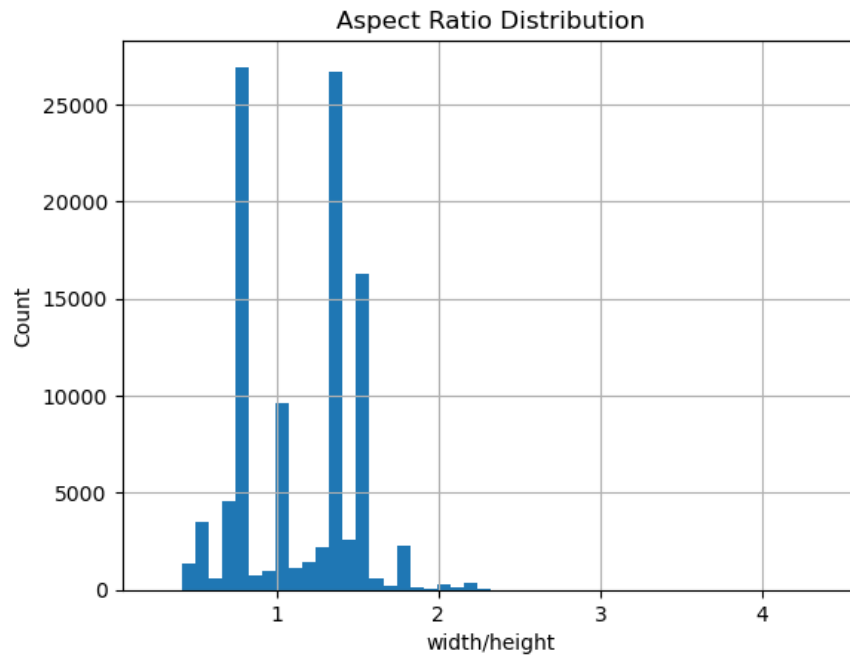


Figure 6: Distribution of Aspect Ratio of Mushrooms

- (b) Only some images have extreme ratios.
- (c) Resizing images before the data input can ensure the consistence.

5. Duplicate Analysis:

- (a) Two types of duplicate analysis were ran: Exact and Near.
- (b) Total of 15,636 (15.2%) images involved in the exact duplicate group.
- (c) Exact-duplicate removal reduce the dataset to 94,698 images, and Near-duplicate removal reduce the dataset to 94,631 images.
- (d) **Exact-duplicate analysis:** Table 2 summarizes the metadata of an example set of exact-duplicate images identified using perceptual hashing. All listed images share the same pHash value (**965a79b4423b7266**) but appear under different file paths and species labels. Figure 7 shows the corresponding visual examples of these images.

Table 2: Metadata of example exact-duplicate images identified using perceptual hashing (pHash = 965a79b4423b7266)

Index	File Path	Class	phash
68621	Classes/poisonous/Amanita_roseotincta/Amanita_roseotincta4.png	poisonous	965a79b4423b7266
73590	Classes/poisonous/Amanita_neoovoidea/Amanita_neoovoidea3.png	poisonous	965a79b4423b7266
73618	Classes/poisonous/Amanita_parvipantherina/Amanita_parvipantherina7.png	poisonous	965a79b4423b7266
73776	Classes/poisonous/Amanita_ibotengutake/Amanita_ibotengutake3.png	poisonous	965a79b4423b7266
73825	Classes/poisonous/Amanita_smithiana/Amanita_smithiana2.png	poisonous	965a79b4423b7266
74007	Classes/poisonous/Amanita_petalinivolva/Amanita_petalinivolva2.png	poisonous	965a79b4423b7266



Figure 7: Corresponding visual examples of the exact-duplicate mushroom images listed in Table 2.

- (e) **Near-duplicate analysis:** Table 3 summarizes the metadata of an example set of near-duplicate images identified using perceptual hashing with a small Hamming distance (**Hamming = 2**). The listed images have similar but not identical pHash values and appear under different file paths. Figure 8 shows the corresponding visual examples of these images.

Table 3: Metadata of example near-duplicate images identified using perceptual hashing with Hamming distance = 2.

Index	File Path	Class	phash
42230	Classes/conditionally_edible/snakeskin_grisette/13.png	conditionally_edible	93cb0de47454ca5b
42236	Classes/conditionally_edible/snakeskin_grisette/6.png	conditionally_edible	93cb0de47454ea4b

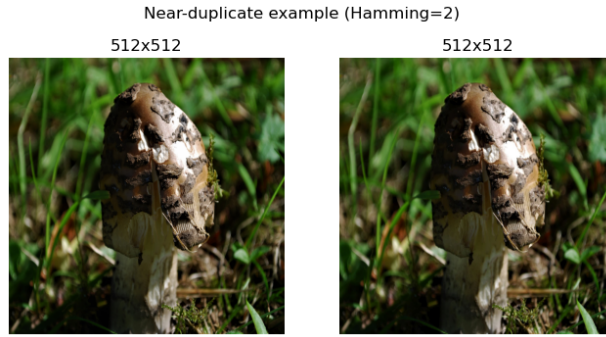


Figure 8: Corresponding visual examples of the near-duplicate mushroom images listed in Table 3.

6. Sample Image Grids:

- (a) Sample image grids were created for each class for a visual sanity check (Figure 9).
- (b) The grids confirmed the class labels and species were assigned in the correct class by showing the visual appearance.

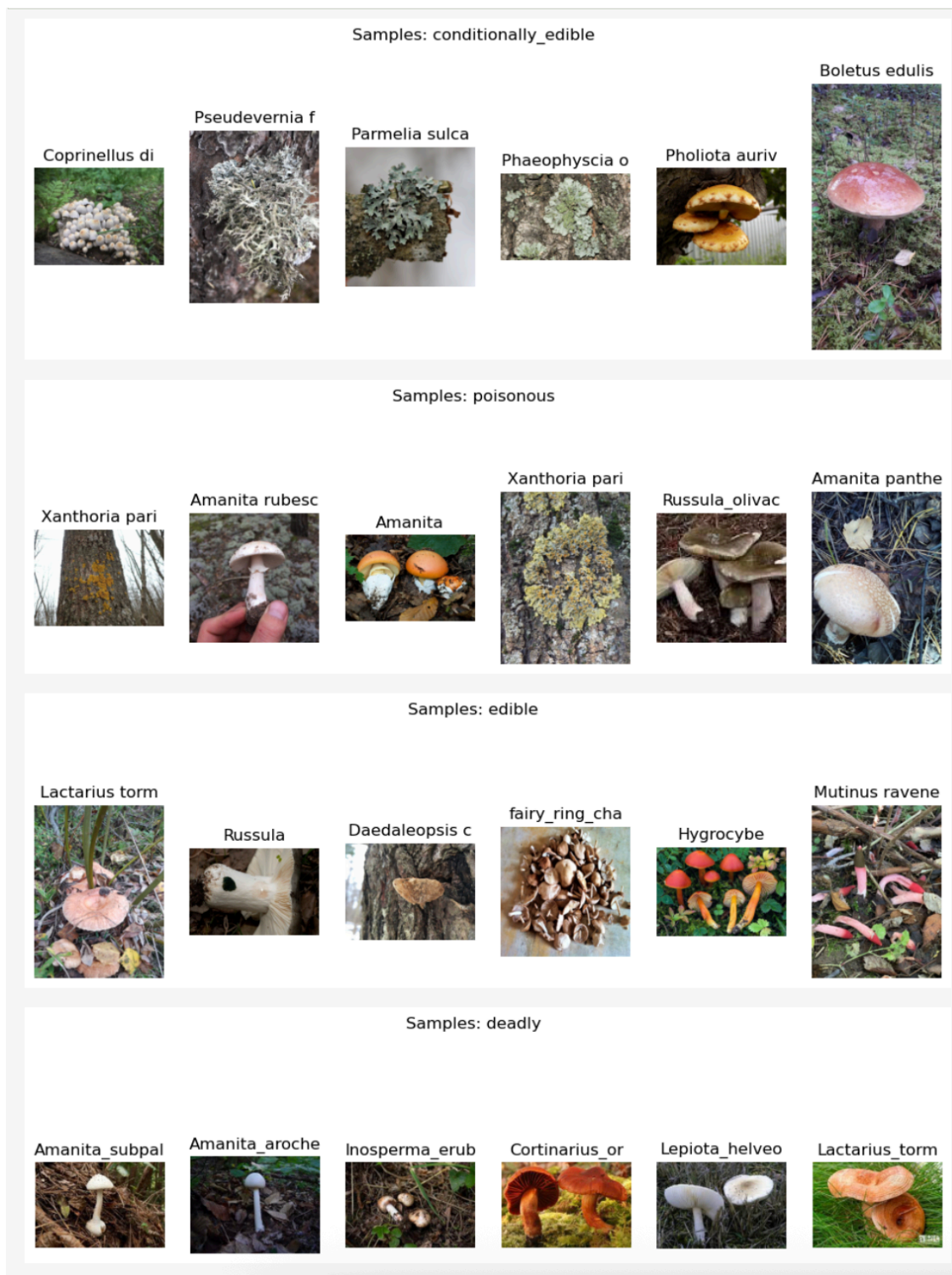


Figure 9: Sample image grids for each mushroom class used for visual sanity checking of class labels and species assignments.

5 Feature Engineering

For our models based on ResNet and Vision Transformer architectures, we adopt a **representation learning** approach rather than manual feature engineering.

These deep learning models are designed to **automatically learn hierarchical feature representations** directly from raw pixel data.

Our role is to prepare the data and guide the learning process so the model can discover the most relevant features.

Possible areas we could investigate:

- **Addressing Data Biases:** Correcting for resolution bias and class imbalance through sampling and augmentation strategies.
- **Providing Robust Data:** Using aggressive data augmentation (e.g., random crops, rotations, color jitter) to teach the model to focus on invariant mushroom traits (cap shape, gill structure) rather than spurious correlations (background, lighting, image quality).
- **Validating Feature Learning:** We will use interpretability tools like Grad-CAM to verify that the model’s learned features align with biologically important regions of the mushroom.

In essence, our “feature engineering” is embedded within the model architecture and training pipeline. This approach enables the model to learn a richer, more robust, and more generalizable set of features than could ever be manually defined for a task as complex and varied as mushroom identification from images.

6 Model Selection and Architecture

We evaluate both convolutional and transformer-based vision models to compare traditional supervised learning with modern attention-based and self-supervised approaches. Mushroom edibility classification depends on subtle visual traits such as cap shape, stem proportions, and gill patterns, which makes it important to capture both local textures and global structural relationships.

6.1 Model Priority and Justification

Our implementation priority is as follows:

1. **ResNet-18:** Serves as our baseline CNN. It is computationally efficient and reliable for image classification, capturing local visual features well.
2. **ResNet-50:** A deeper convolutional network designed to learn more detailed hierarchical features. Comparing ResNet-50 to ResNet-18 allows us to assess the impact of increased depth and complexity.

3. **Swin Transformer Tiny:** A supervised transformer model that captures both local and global image relationships using hierarchical shifted windows. It is particularly suitable for structured visual differences that CNNs may miss.
4. **DINO-pretrained Vision Transformer (ViT-Small):** A self-supervised model trained without labels using a student–teacher framework. Fine-tuning this model allows us to evaluate whether self-supervised representations improve generalization and robustness. This is a lower-priority stretch goal.

6.2 Model Architectures

ResNet-18 and ResNet-50 are based on deep residual learning, which enables effective training of deep convolutional networks through skip connections that stabilize gradients [He et al. \(2015\)](#). ResNet-18 is smaller and simpler, providing a fast baseline, while ResNet-50 uses bottleneck blocks and greater depth to learn more complex hierarchical visual patterns.

Vision Transformers (ViT) divide images into patches and apply self-attention to model relationships across image regions [Dosovitskiy et al. \(2021\)](#). Swin Transformers extend this approach by using hierarchical shifted windows to efficiently capture both local and global features [Liu et al. \(2021\)](#). The DINO framework is a self-supervised learning approach for Vision Transformers that generates meaningful visual representations without labels using a student–teacher training strategy [Caron et al. \(2021\)](#).

6.3 New Architecture Experiments

If time allows, we will experiment with fine-tuning the DINO-pretrained Vision Transformer rather than training a self-supervised model from scratch. This will allow us to test the impact of self-supervised pretraining on downstream mushroom classification. Additionally, we plan to explore uncertainty estimation techniques such as Monte Carlo dropout and small deep ensembles, as well as a reject option for low-confidence predictions, which is important for safety-critical tasks like mushroom edibility classification.

7 Progress Report Details

- The metadata was consolidated into a single CSV file.
- Mushroom data were de-duplicated, leaving only unique mushroom images.
- Performed EDA and generated useful insights on image dimensions, distributions, and resolutions.
- Researched different models and finalized three models for implementation, with one additional model planned if time allows.
- The processed data are ready to be used for model training.

8 Team Member Contribution

- **JoJo:**
 - Worked on Section 1 **Background**, partial of Section 3.2 **Pipeline design and details** (3.2.3-3.2.5), Section 3.3 **Description of outputs** and Section 4 **EDA Description**
 - Implemented and validated exact- and near-duplicate detection results in data pipeline and EDA.
 - Designed data standardization and data augmentation.
 - Conducted EDA and generated visualization for basic quality checks, class distribution analysis, image dimension distribution, aspect ratio analysis, duplicate analysis, and visual sanity checks.
 - Reviewed and revised the full report.
 - Actively participated in team discussions.
- **Xuewen:**
 - Consolidated image datasets organized in a directory hierarchy into a unified CSV file with metadata of each image (e.g. paths, dimensions, resolutions).
 - Partially worked on Section 3.2 **Pipeline design and details**.
 - Led Section 7 **Progress Report Details**.
 - Coordinated the creation of images and tables and ensured proper referencing throughout the report.
 - Reviewed and revised the full report.
 - Actively participated in team discussions.
- **Nicholas:**
 - Developed and documented the Section 6 **Risks and Mitigation**, including CNNs, transformer architectures, and self-supervised approaches.
 - Drafted and finalized the Section 9 **Risks and Mitigation**, identifying potential project challenges and proposed solutions.
 - Contributed to team discussions on model architecture, experimental design, and project priorities.
- **Joanna:**
 - Perform EDA on the dataset - explore/understand the dataset, general statistics for initial analysis, etc.
 - Check duplicates on the Mushroom Dataset.
 - Removes duplication, save the clean dataset in CSV file for future model use.
 - Worked on Section 5 **Feature Engineering**.

9 Risks and Mitigation

Several risks could affect the successful completion of our mushroom classification project. The first risk is the variable directory depth in the dataset, where some images are stored

at depth 3 and others at depth 4. This inconsistency could lead to incorrect labeling of species and classes. We mitigate this risk by implementing conditional logic in the data pipeline to handle variable directory depths, performing thorough exploratory data analysis (EDA) using `mushroom_paths.csv`, and applying data augmentation for underrepresented classes if necessary.

The second risk concerns class imbalance and model performance. While the dataset contains over 102,000 images, certain classes may be underrepresented, which could lead to overfitting, especially in deeper models like ResNet-50 or transformer-based models. We address this by using regularization techniques such as dropout and weight decay, applying stratified train-test splits, monitoring per-class performance, and implementing weighted cross-entropy loss to give more importance to minority classes and improve model robustness.

The third risk is computational resources. Training multiple models, including deeper CNNs (ResNet-50) and transformer models (Swin Transformer Tiny), can be resource-intensive. We mitigate this by following our prioritized model order (ResNet-18, ResNet-50, Swin Transformer Tiny) and selecting smaller model variants, using efficient fine-tuning strategies such as learning rate warmup and gradient accumulation, and only experimenting with DINO-pretrained Vision Transformers if resources and time permit.

The final risk is safety-critical: misclassifying poisonous mushrooms as edible could have severe real-world consequences. Standard accuracy metrics may not fully capture this risk. To reduce potential harm, we will prioritize predictions with high confidence and focus on achieving strong recall for poisonous classes, ensuring the model is more likely to identify toxic mushrooms correctly. We will also use detailed confusion matrix analyses to monitor and understand specific misclassification patterns between visually similar species.

References

- Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin.** 2021. “Emerging Properties in Self-Supervised Vision Transformers.” [\[Link\]](#)
- Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby.** 2021. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.” [\[Link\]](#)
- He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun.** 2015. “Deep Residual Learning for Image Recognition.” [\[Link\]](#)
- Liu, Ze, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo.** 2021. “Swin Transformer: Hierarchical Vision Transformer using Shifted Windows.” [\[Link\]](#)